



BRAC UNIVERSITY
Department of Computer Science Engineering
Mid Term Examination, Spring 2024
CSE 350: Digital Electronics and Pulse Techniques



Total Marks: **60 (20 x 3)**
Time: **90 Minutes**

- Total no. of questions are **four (4)**. Answer any **three (3)**.
- Figure in bracket [] next to each question indicates marks for that question.

Name:	ID:	Section:
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Question 1 **[20]**

For the RTL circuit in Figure 1, assume $V_{\gamma}(\text{transistor}) = 0.5 \text{ V}$, $\beta_F = 25$, $V_{BE}(\text{saturation}) = 0.8 \text{ V}$, and $V_{CE}(\text{saturation}) = 0.2 \text{ V}$.

CO2	(a)	Determine the value of V_R such that when $V_{in} = 0.2 \text{ V}$, the power dissipation is 1 mW. [Assume $T1$ is in cutoff]	[7]
CO2	(b)	Calculate the maximum value of V_{in} that can be applied such that transistor $T1$ is still in cutoff state. [Assume $V_R = -12 \text{ V}$ for this question]	[7]
CO2	(c)	Determine the minimum value of β_F such that the logic gate does not malfunction. [Assume $V_R = -12 \text{ V}$ for this question]	[6]

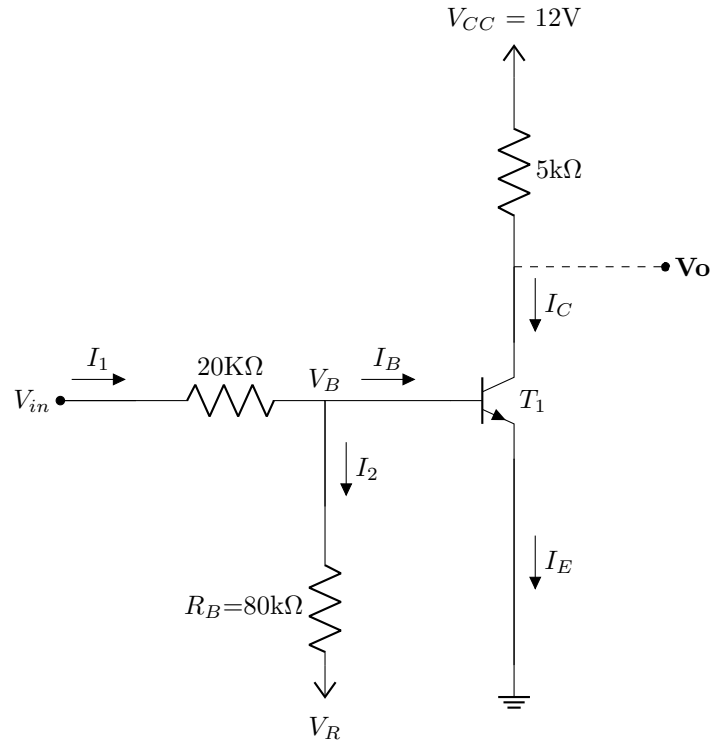


Figure 1

Question 2

[20]

Blocks A and B are two different circuits, shown in Fig 2a and 2b, respectively. The final circuit is constructed using these blocks and additional diodes shown in Figure 3, where V_{X1} , V_{X2} , V_{X3} and V_{X4} are the inputs of the two A blocks. Consider the following two input combinations:

Case 1: $V_{X1} = 0V$, $V_{X2} = 0V$, $V_{X3} = 0V$, $V_{X4} = 0V$

Case 2: $V_{X1} = 5V$, $V_{X2} = 5V$, $V_{X3} = 5V$, $V_{X4} = 5V$

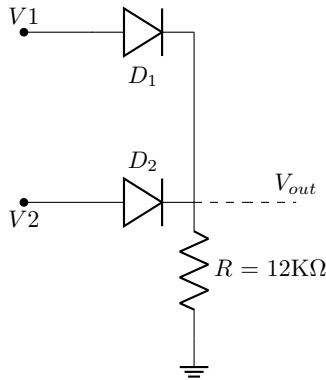


Fig 2a: Block A

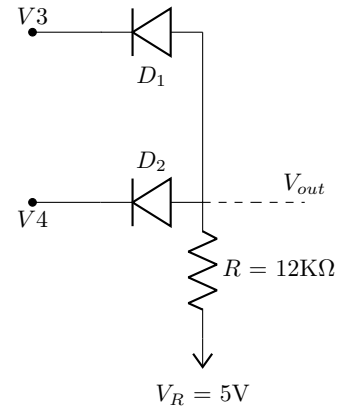


Fig 2b: Block B

Figure 2

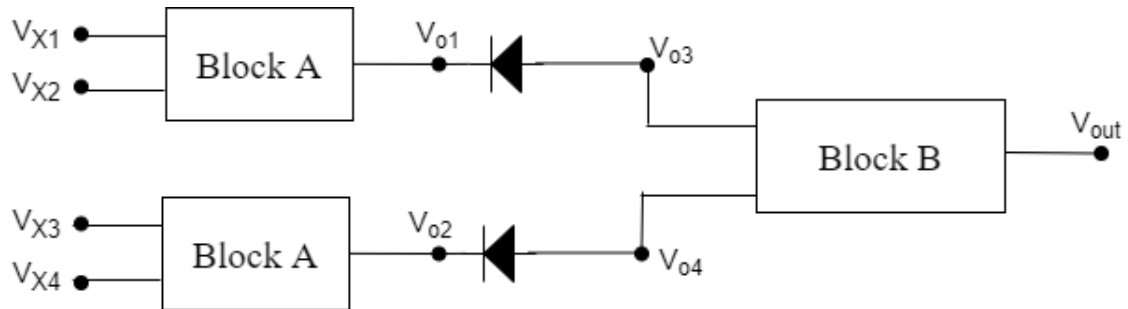


Figure 3

CO1	(a)	Find the voltages at Vo1, Vo2, Vo3, Vo4 and Vout for case 2	[4]
CO1	(b)	Calculate the total power dissipation by each of the resistors in the overall circuit for case 1. [Hint: to calculate power dissipated by each resistor, calculate the current through that resistor and use formula $P = I^2 R$]	[12]
CO1	(c)	Implement the following Boolean function using only Block A and Block B. $V_{out} = ((V_{X1} + V_{X2}) \cdot V_{X3} \cdot V_{X4})$. (Only draw the circuit using the blocks)	[4]

Question 3

[20]

In the 2-input DTL NAND Logic shown below, the input voltages V_a and V_b can be either 0.2V (Low) or 8V(High). The utilized diodes and transistor have **different operating voltages than usual**. Important parameters are given below:

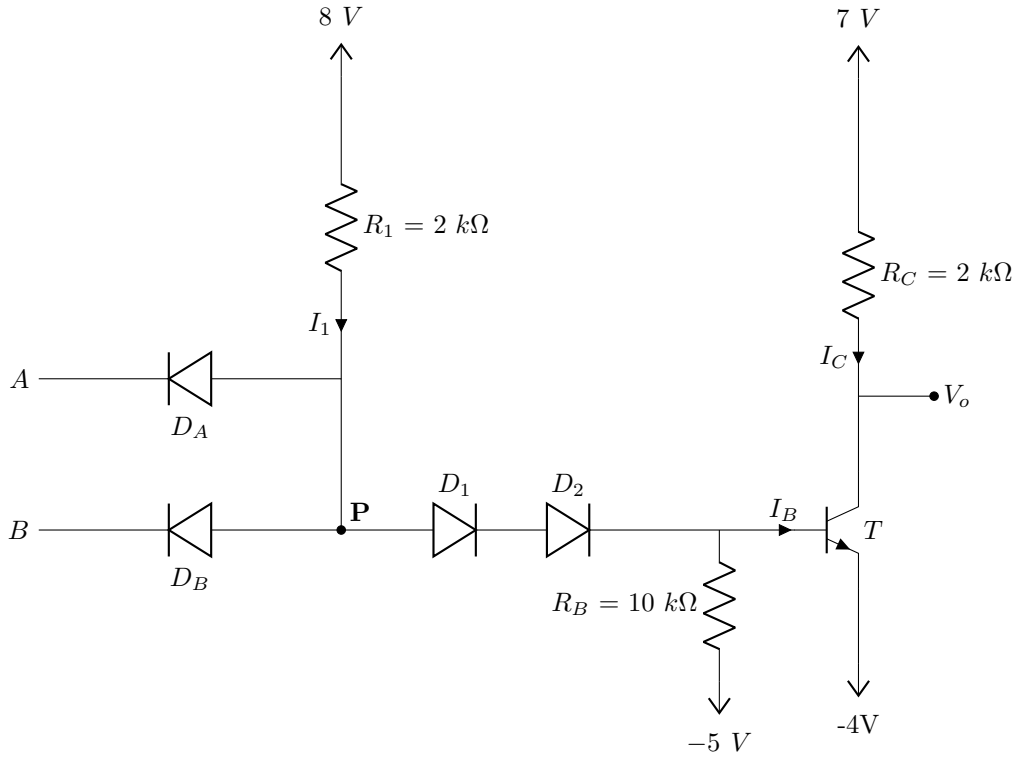


Figure 4

Table 1: Table for Question 3

β_F	6	$V_{BE}(Sat)$	1V	$V_{CE}(Sat)$	0.5V
$V_\gamma(BJT)$	0.8V	$V_\gamma(Diode)$	0.6V	$V_d(Diode)$	0.7V

CO1	(a)	Determine the minimum value of V_p for the transistor to be just on .	[4]
CO1	(b)	Determine V_o , I_1 , I_B and I_C if both the inputs are in high state.	[7]
CO1	(c)	Is the given β_F suitable for the circuit? [Hint: Calculate β_{min}]	[2]
CO1	(d)	Calculate the maximum dissipated power achievable from the circuit?	[6]

Question 4

[20]

For the given TTL circuit in Figure 5, **common emitter current gain**, $\beta_F = 30$, $V_{BE(sat)} = 0.9V$, $V_{CE(sat)} = 0.1V$, and **reverse common emitter current gain** of the transistors $\beta_R = 0.1$. Assume both inputs are **logical High** (i.e. $5V$).

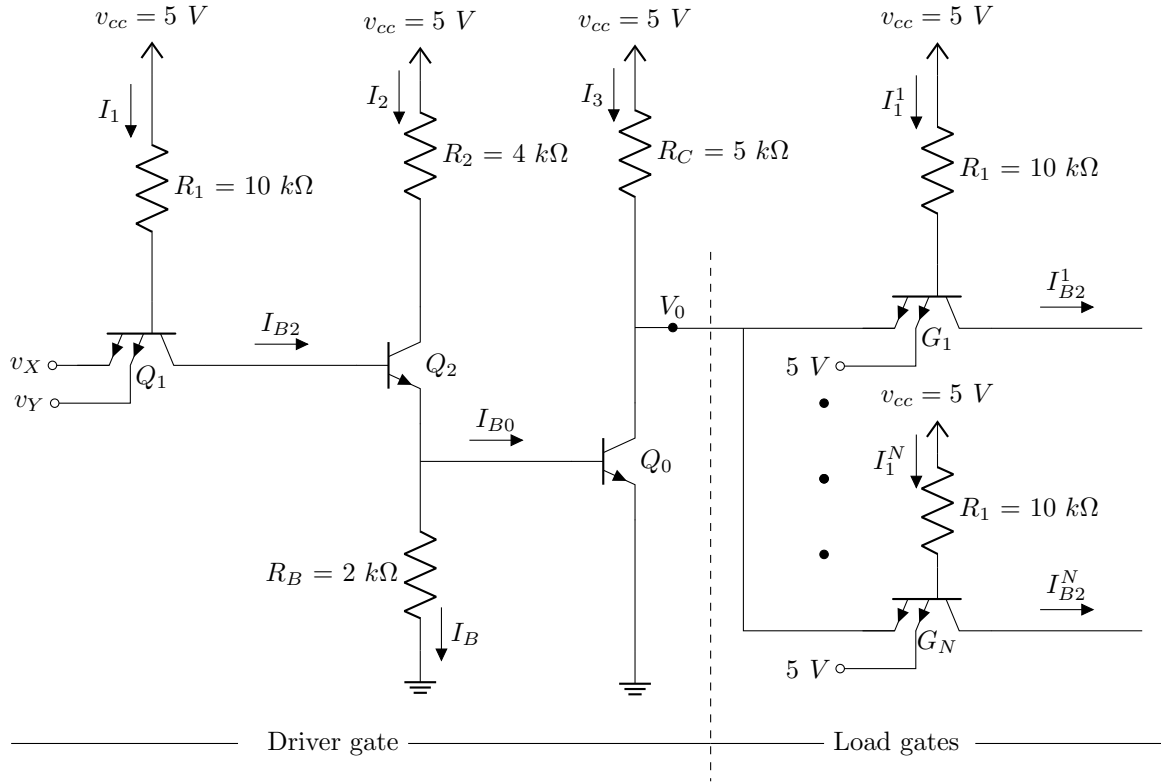


Figure 5

CO2	(a)	Determine the state of all the transistors on the driver side and Prove the state of the Q_0 transistor when the load circuits are disconnected from the driver. [<i>Only for the given input condition</i>].	[5]
CO2	(b)	Calculate the fanout for this input configuration once the loads are connected as shown in Figure 5.	[10]
CO2	(c)	If the loadside input terminal, V_y (the terminal not connected to the driver), is changed from $5V$ to $0V$, explain with reason the amount of increase/decrease of the fanout calculated in part (b).	[5]